

Chapter 8

Understanding Markets and Industry Changes

Learning Objectives

- 8.1 Know how to define the market before using supply-demand analysis.
- 8.2 Describe buyer behavior and seller behavior in a competitive market.
- 8.3 Explain the demand and supply curves and how they relate to the market equilibrium.
- 8.4 Describe the changes that occur at the industry level by using supply and demand curves.

Chapter 8 – Main Points

- A market has a product, geographic, and time dimension. Define the market before using supply–demand analysis.
- *Market demand* describes buyer behavior; *market supply* describes seller behavior in a competitive market.
- If price changes, *quantity demanded* increases or decreases (represented by a **movement along the demand curve**).
- If a factor other than price (like income) changes, we say that demand curve increases or decreases (a **shift of demand curve**).
- **Supply curves** describe the behavior of sellers and tell you how much will be sold at a given price.
- **Market equilibrium** is the price at which quantity supplied equals quantity demanded. If price is above the equilibrium price, there are too many sellers, forcing price down, and *vice versa*.
- Prices convey valuable information; high prices tell buyers to conserve and sellers to increase supply..
- Making a market is costly, and competition between market makers forces the bid–ask spread down to the costs of making a market. If the costs of making a market are large, then the equilibrium price may be better viewed as a spread rather than a single price.

Warm-Up

- Have you ever stopped buying something because of a small price increase? Why?

Defining the Market

- A market exists at the intersection of:
 - Product – What is being bought or sold?
 - Geography – Where does the transaction occur?
 - Time – When is the exchange taking place?
- Market definitions must be specific to be useful.
- Poorly defined markets lead to bad analysis and wrong conclusions.
- Always define the market before applying supply–demand tools.

Demand

- Demand is a relationship that economist use to describe the relationship between what consumers are willing to buy and are able to buy.
- Without the willingness and ability to buy a product, a consumer or group of consumers have no demand for the product in question.
- Economists and market analyst form demand relationships when a set of prices and corresponding quantities are paired.
- This pairing can be found in a demand schedule which is a table showing unique price and quantity pairings.

Demand Schedule

Let's examine market data for the iPhone in the following demand schedule:

Price	Quantity Demanded
\$699	0
599	10
499	20
399	30
299	40
199	50
99	60

Notice at a price of \$699 quantity demanded is 0. There may be consumers who are willing to pay for the good but are not able. Or there may be consumers who are able to pay for the good but not willing. Or those who are not willing or able.

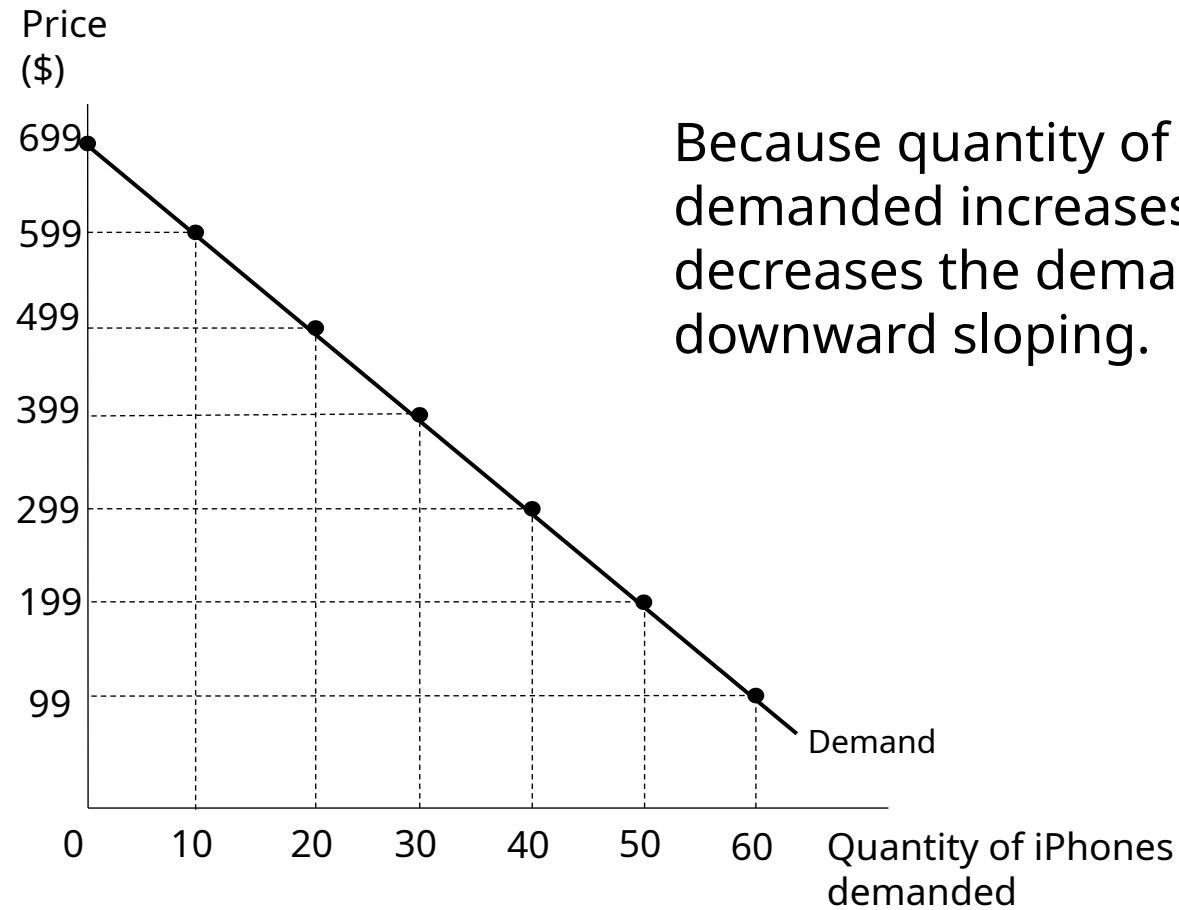
As the price drops quantity demanded increases, up to 60 iPhones at a price of \$99.

Law of Demand -There is an inverse relationship between the price of a good and the quantity demanded of the good, *ceteris paribus*.

Ceteris paribus means other things held constant

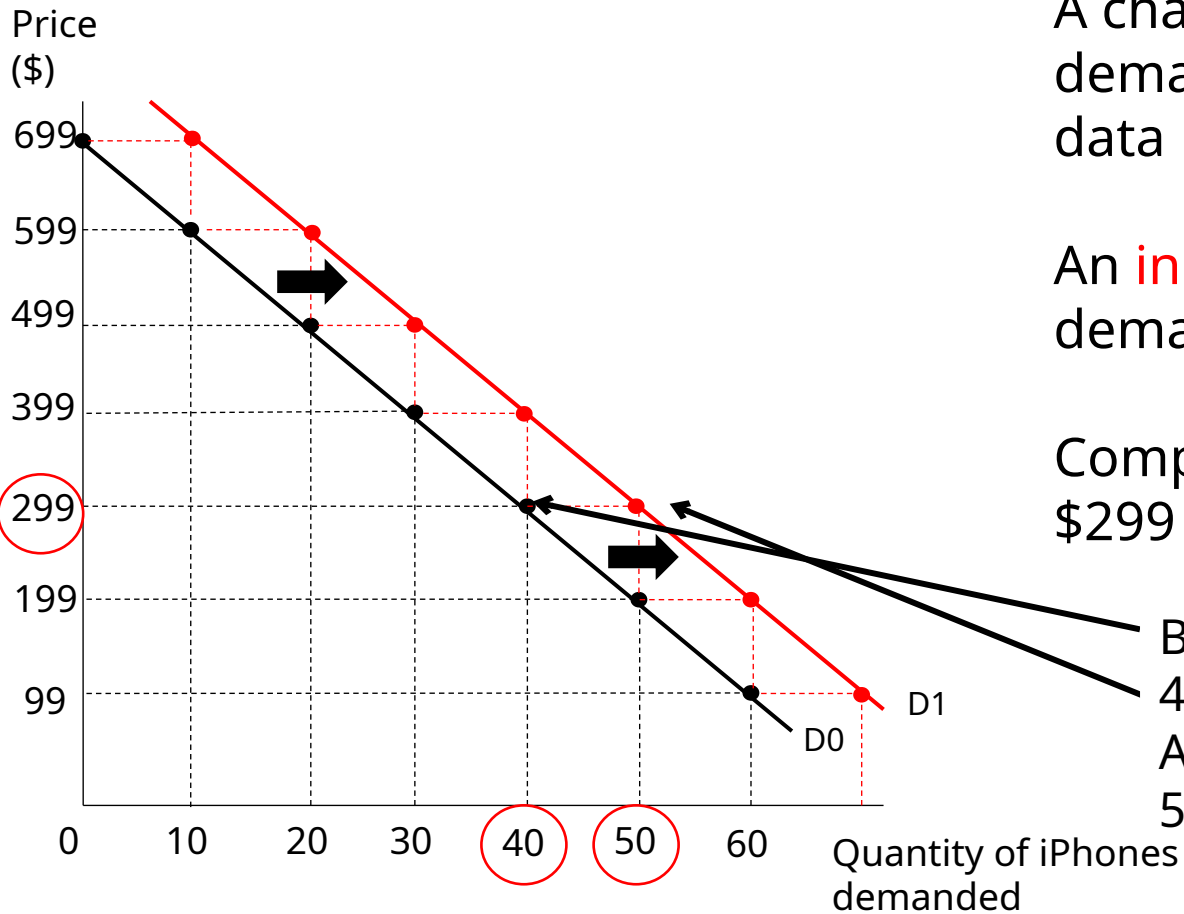
The Demand Curve

Price	Quantity Demanded
\$699	0
599	10
499	20
399	30
299	40
199	50
99	60



Because quantity of iPhones demanded increases as price decreases the demand curve is downward sloping.

Shifting the Demand Curve



A change in one of the determinants of demand causes a change in each quantity data point.

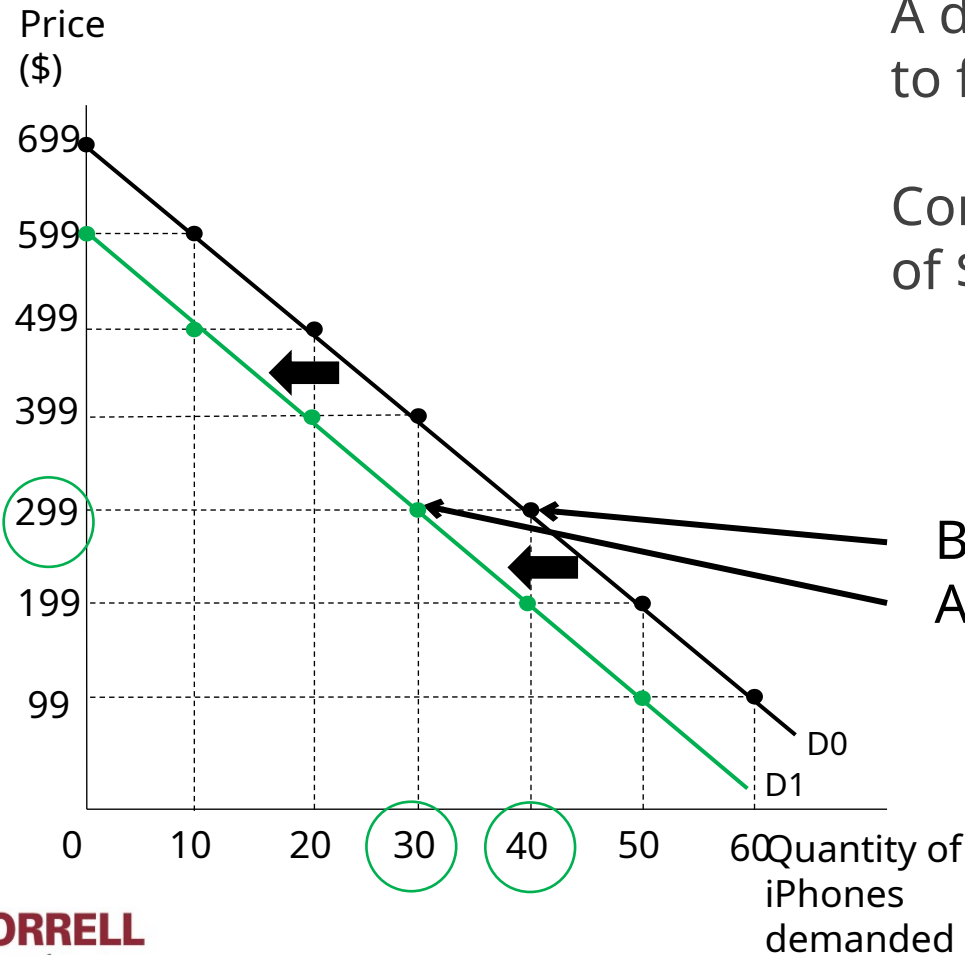
An **increase in demand** will cause quantity demanded to increase at each previous price.

Compare the quantity demanded at a price of \$299

Before on demand curve D0: $Q = 40$

After on demand curve D1: $Q = 50$

Shifting the Demand Curve



A decrease in demand will cause quantity to fall at each previous price.

Compare the quantity demanded at a price of \$299

Before on demand curve D0: $Q = 40$

After on demand curve D1: $Q = 30$

Factors that Shift Demand

- There are several factors that shift demand
- Income
 - An **increase** (**decrease**) in income **increases** (**decreases**) demand
 - Normal vs. Inferior Goods
- Tastes/Preferences
 - An **increase** (**decrease**) in tastes/preferences **increases** (**decreases**) demand
- Population
 - An **increase** (**decrease**) in population **increases** (**decreases**) demand

Factors that Shift Demand

- Price of another good
 - **Complements** – two goods that are consumed together. An **increase** in the price of one good **decreases** the demand for the other
 - **Substitutes** – two goods that can be switched for one another. An **increase** in the price of one good **increases** the demand for the other
- Expectations
 - Price – If we expect a higher price tomorrow, we demand more today
 - Income – If we expect a higher income tomorrow, we demand more today.

Case Study: Cash for Clunkers

- Cash for Clunkers was a government program designed to help stimulate the economy.
- The problem with this program is that all it really did was shift consumer demand forward meaning that sales that would have occurred a few months in the future occurred today.
- Moreover, the falling vehicle demand was never really addressed by the program. Essentially a Band-Aid was placed on a cutoff limb.
- Another issue with this program was the increase in used car prices that never would have occurred.
 - In essence, clunkers were traded in eroding the junk car market quantity supply. This in turn caused fewer junk cars to be available for purchase, thus driving up those prices.
 - Who was ultimately hurt? One of the groups the program intended to help—the poor.
 - Sources: [Managerial Econ blog](#), [Mian and Sufi \(2010\)](#)

Supply

- Supply is the relationship between the price of a good and the amount of a good the producer is willing and able to supply
- Like demand, there has to be a willingness and ability to sell the product, without it there will be no supply.
- Quantity supplied comes from a production process where the factors of production (land, labor, capital and entrepreneurial ability) are combined to create a product to be sold in the market.
- The unique pairing of quantity supplied, and price can be put into a table or supply schedule.

Supply Schedule

Below is the supply schedule for the iPhone:

Price	Quantity Supplied
\$699	60
599	50
499	40
399	30
299	20
199	10
99	0

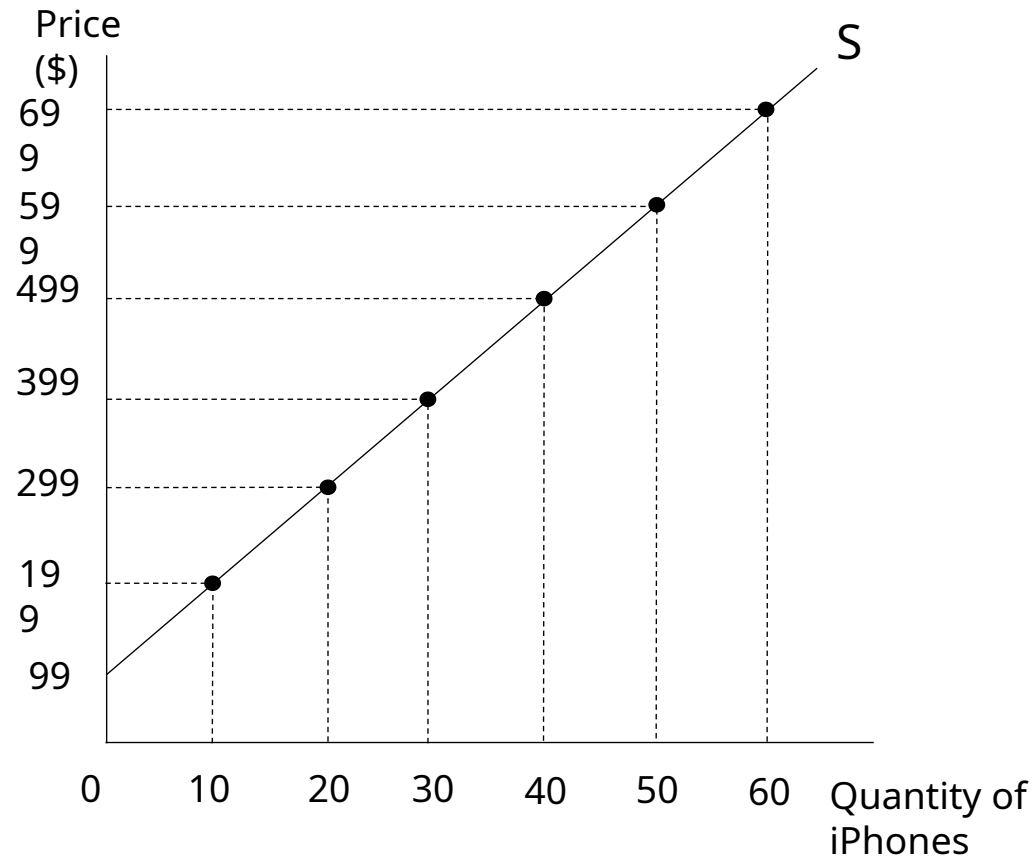
At a price of \$699, the producer is willing and able to sell 60 iPhones.

As the market price falls from \$699 to \$99, quantity supplied does as well (from 60 to 0).

Law of Supply – there is a direct relationship between quantity supplied and price, ceteris paribus.

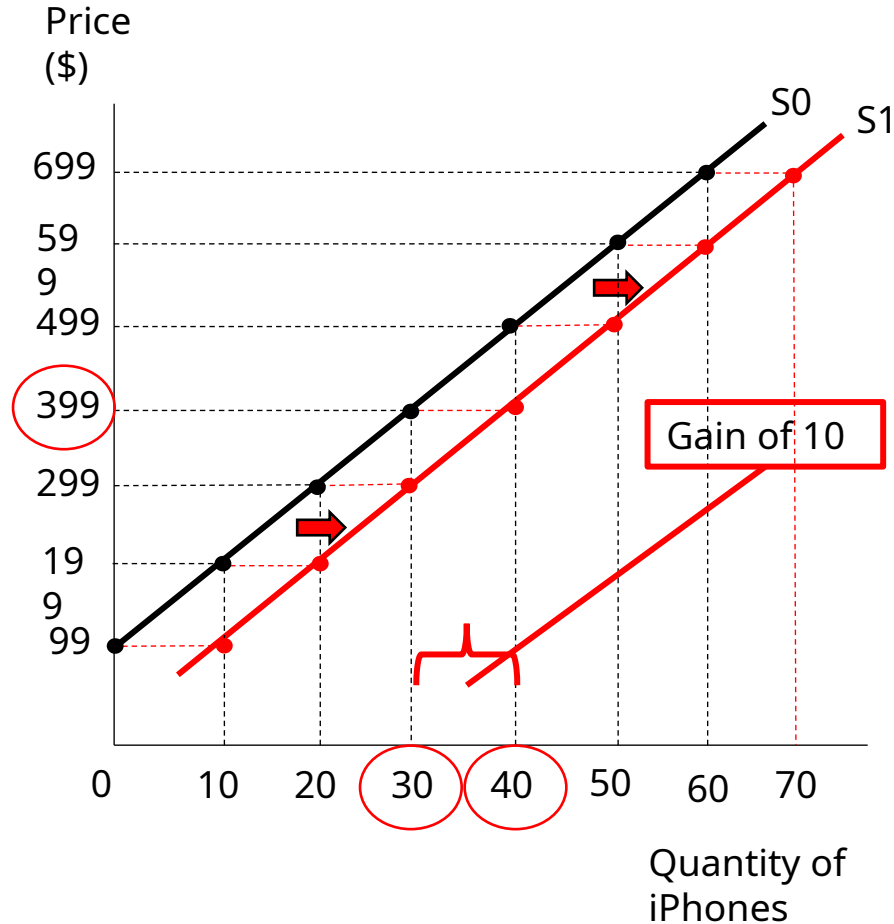
Constructing a Supply Curve

Price	Quantity Supplied
\$699	60
599	50
499	40
399	30
299	20
199	10
99	0



Each price and quantity point are matched on their respective axis. Because quantity increases as price increases, supply is positively sloped

Shifts in the Supply Curve



An **increase** in the supply curve causes the entire curve to shift to the right

At each of the previous prices, quantity supplied has increased from its original values.

For example, at a price of \$399

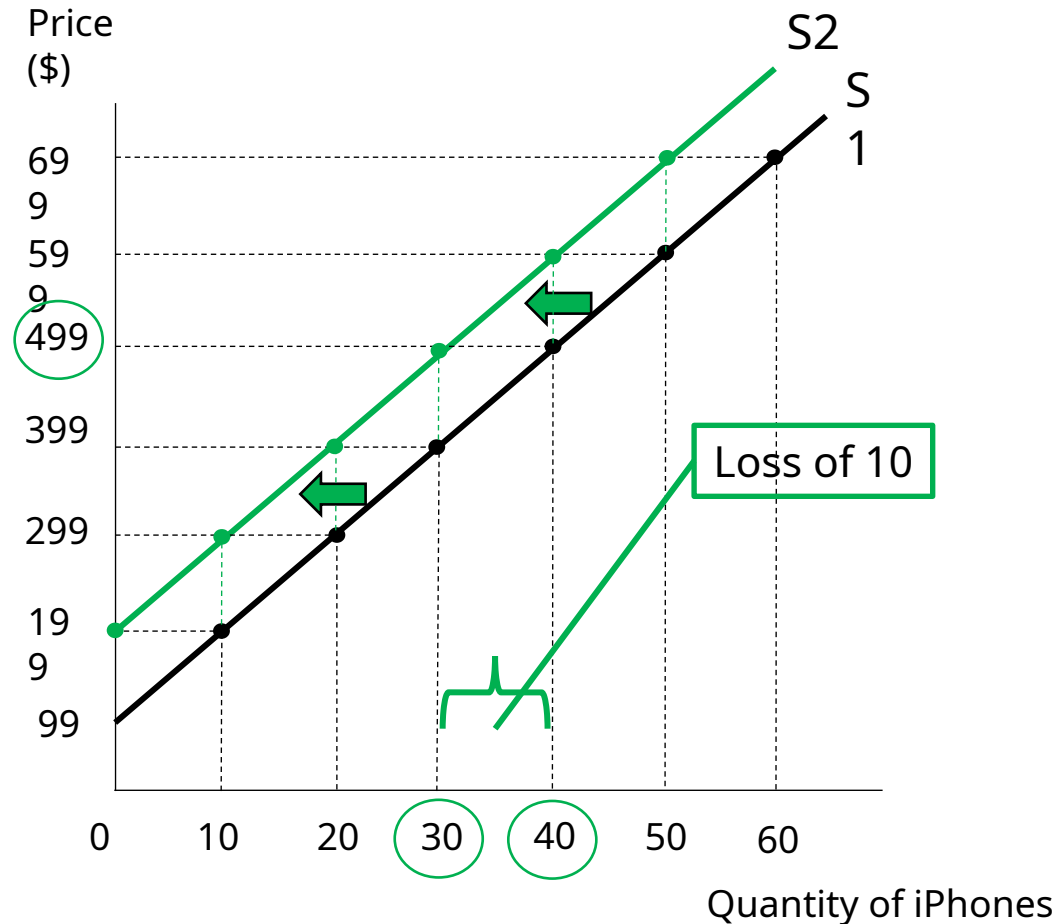
Before

Quantity supplied = 30

After

Quantity supplied = 40

Shifts in the Supply Curve



A **decrease** in the supply curve causes the entire curve to shift to the left.

At each previous price, quantity supplied has been reduced.

For example, at a price of \$499:

Before
Quantity = 40

After
Quantity = 30

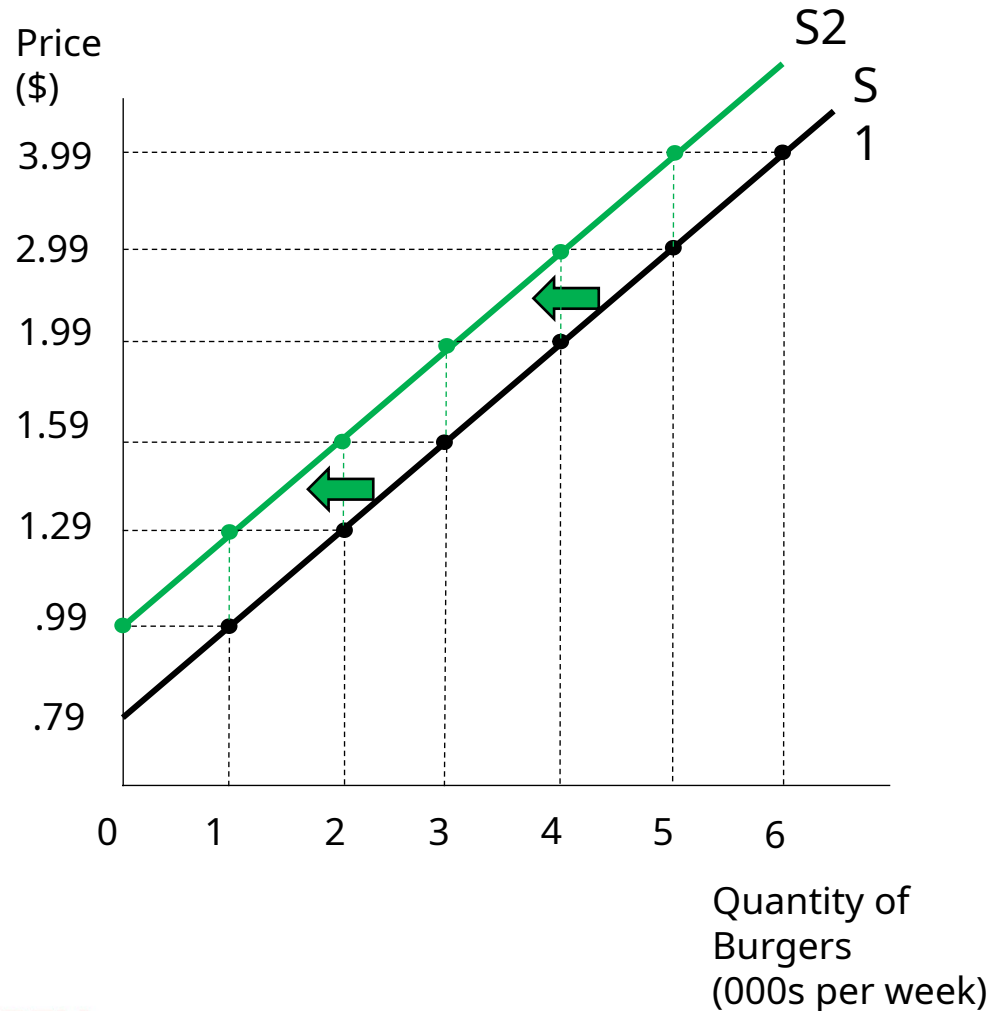
Factors that Shift Supply

- Several factors shift supply:
- Number of producers or firms
 - An **increase** (**decrease**) in the number of producers, **increases** (**decreases**) supply
- Technology
 - An **increase** (**decrease**) in technology, **increases** (**decreases**) supply
- Price of resources
 - An **increase** (**decrease**) in the price of a resource, **decreases** (**increases**) supply

Factors that Shift Supply

- Taxes and Subsidies
 - An **increase** (**decrease**) in taxes, **decreases** (**increases**) supply
 - An **increase** (**decrease**) in subsidies, **increases** (**decreases**) supply
- Price Expectations
 - Higher (lower) prices in the future, lead to less (more) supply today
- Prices of Related Goods
 - An increase in the price of a substitute in production, **decreases** the supply of the other good
 - An increase in the price of a complement in production, **increases** the supply of the other good

An Increase/Decrease in Wages

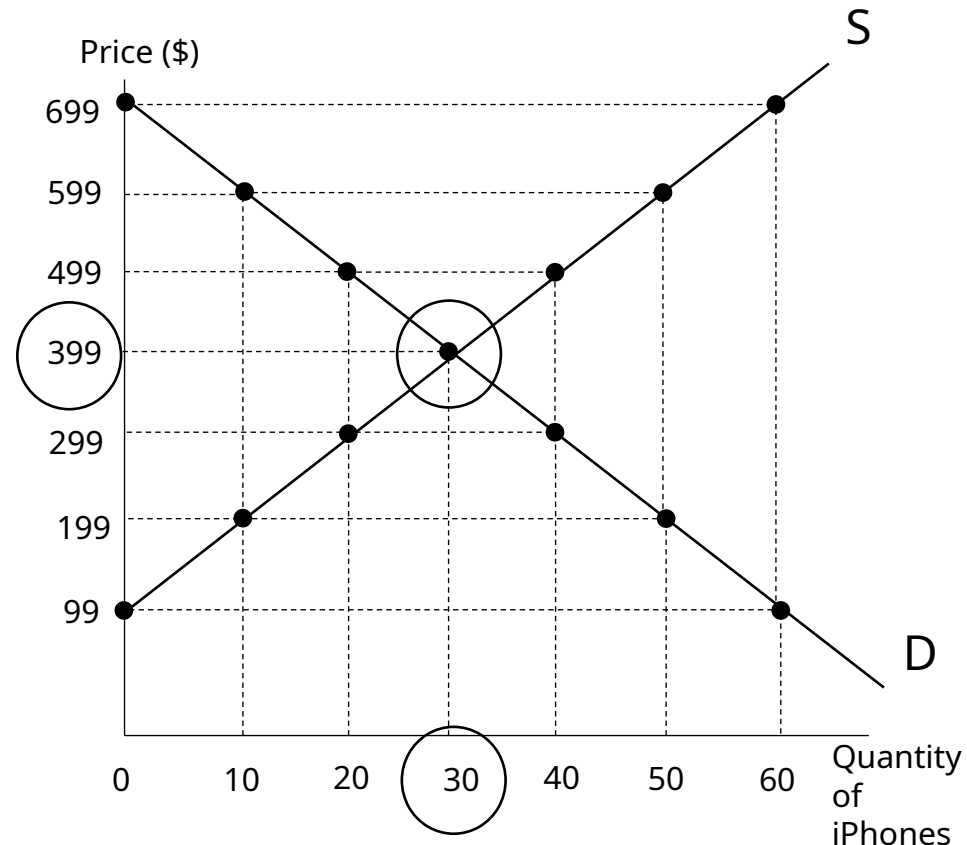


Demand and Supply

Where the demand and supply curves intersect, we say **market equilibrium** has been established. Below the market equilibrium is found at a price of \$399 and 30 iPhones.

Market equilibrium is also found in the demand and supply schedule where quantity demanded equals quantity supplied

Price	Quantity Demanded	Quantity Supplied
\$699	0	60
599	10	50
499	20	40
399	30	30
299	40	20
199	50	10
99	60	0



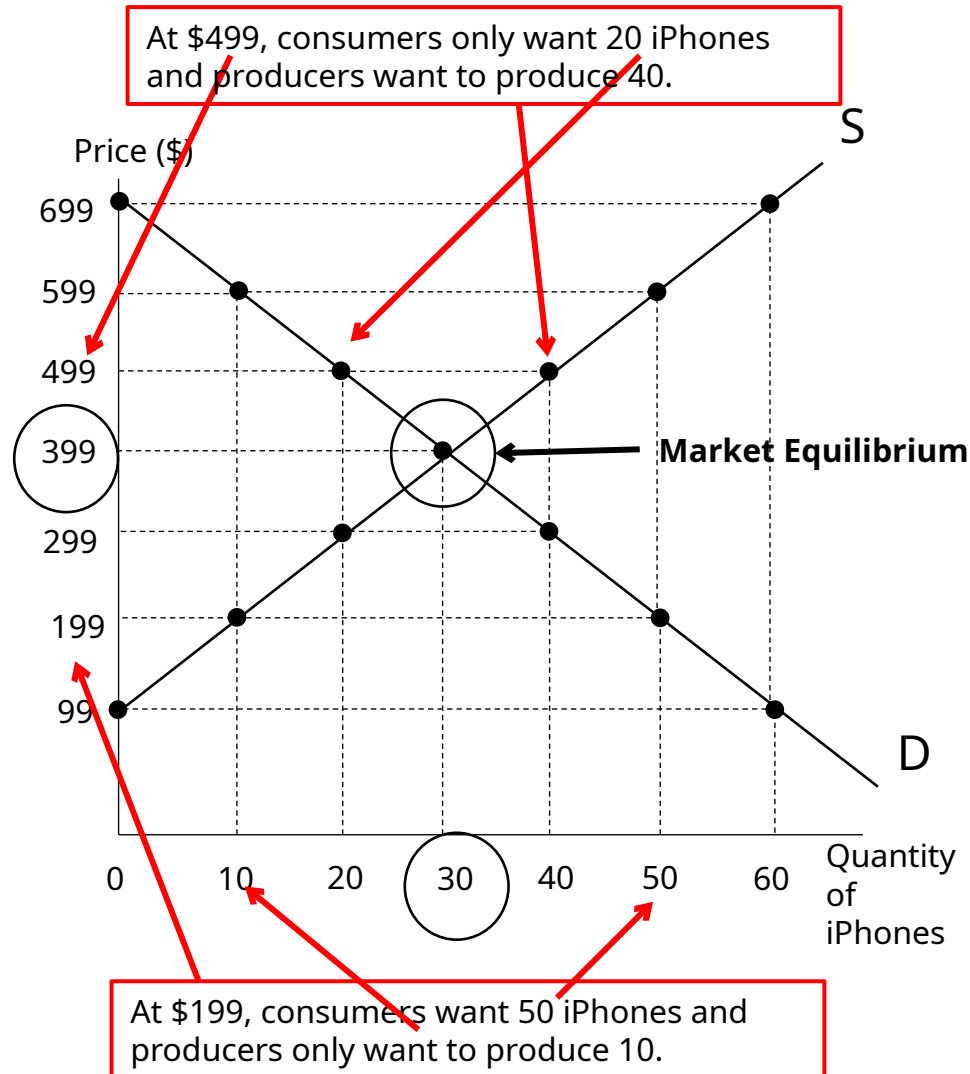
Market Equilibrium

Equilibrium is the condition where price regulates how much is bought and sold.

Consumers want to buy 30 iPhones at a price of \$399, and producers want to sell 30 iPhones at a price of \$399.

Any amount above the equilibrium price leaves consumers wanting less and producers wanting to sell more.

Any amount below the equilibrium price leaves producers wanting to sell less and consumers wanting to buy more.



Excess Demand and Excess Supply

The market is in equilibrium when quantity demanded equals quantity supplied. At a price of \$399, we find market equilibrium because quantity demanded equals quantity supplied.

If price is higher than \$399 then the quantity supplied is greater than quantity demanded. This is called a **surplus**

If price is lower than \$399 then the quantity demanded is greater than quantity supplied. This is called a **shortage**

Price	Quantity Demanded		Quantity Supplied	Surplus (-) Shortage (+)
\$699	0	-	60	-60
599	10	-	50	-40
499	20	-	40	-20
399	30	-	30	0
299	40	-	20	20
199	50	-	10	40
99	60		0	60

Remember the willingness and ability to pay or produce principle. At a price of \$599, only 10 iPhones are bought. This means consumers were only willing and able to purchase 10 iPhones at a price of \$599.

At a price of \$599 producers are willing to sell 50 iPhones but are only able to sell 10. These actions buy consumers and producers create the surplus of iPhones.

Surpluses and Shortages

Surpluses and shortages can be seen on the demand and supply graph as well.

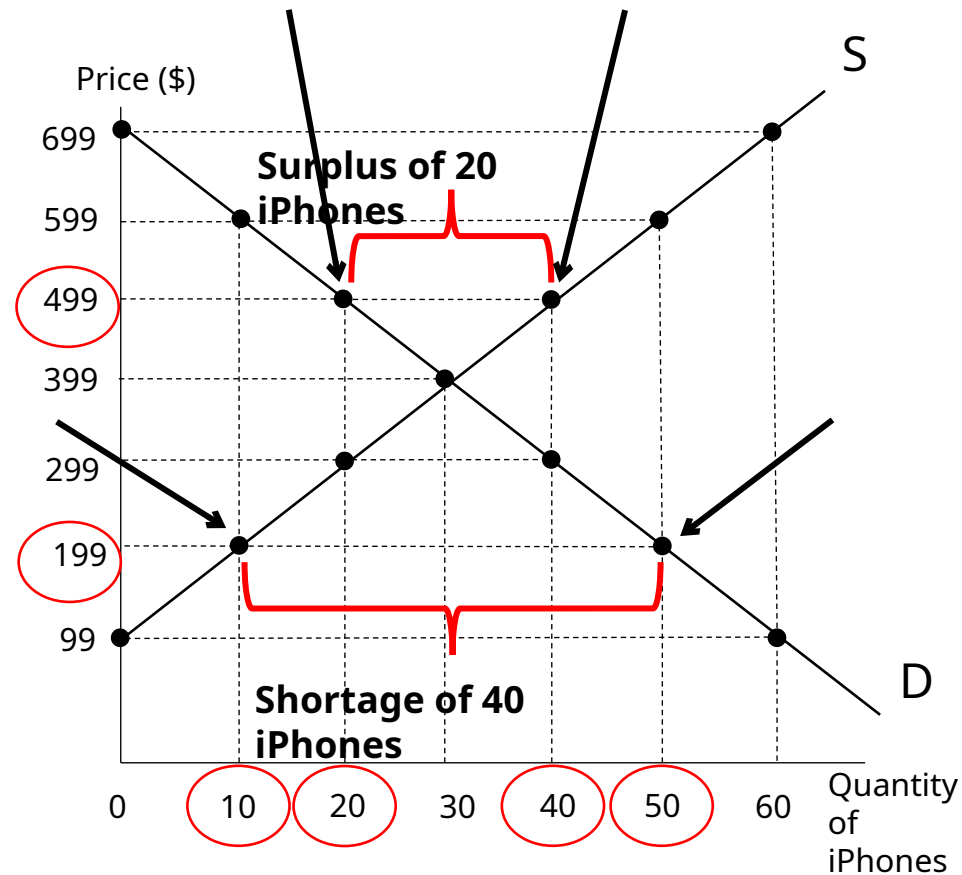
When the price is greater than equilibrium price a surplus occurs

For example, at a price of \$499 there is a surplus of 20 iPhones

When the price is below the equilibrium price a shortage occurs

For example, at a price of \$199 there is a shortage of 40 iPhones

Note: surpluses occur above market equilibrium, shortages occur below market equilibrium



Market Equilibrium

- For a given market we have the following demand and supply functions:

$$Q_D = 100 - 5P$$

$$Q_S = 10 + 10P$$

Determine the inverse demand and supply functions. Find the equilibrium price and quantity. If the price were \$4 , would there be a surplus or shortage?

Determine the inverse demand and supply functions.

$$Q_D = 100 - 5P$$

$$-100 + Q_D = -5P$$

$$20 - 0.2Q_D = P$$

$$P = 20 - 0.2Q_D$$

$$Q_S = 10 + 10P$$

$$-10 + Q_S = 10P$$

$$-1 + 0.1Q_S = P$$

$$P = -1 + 0.1Q_S$$

Find the equilibrium price and quantity.

$$Q_D = 100 - 5P$$

$$Q_S = 10 + 10P$$

$$100 - 5P = 10 + 10P$$

$$90 = 15P$$

$$6 = P$$

$$Q_D = 100 - 5P$$

$$Q_D = 100 - 5(6)$$

$$Q_D = 70$$

$$Q_S = 10 + 10P$$

$$Q_S = 10 + 10(6)$$

$$Q_S = 70$$

If the price were \$4 , would there be a surplus or shortage?

$$Q_D = 100 - 5P$$

$$Q_S = 10 + 10P$$

$$Q_D = 100 - 5(4)$$

$$Q_S = 10 + 10(4)$$

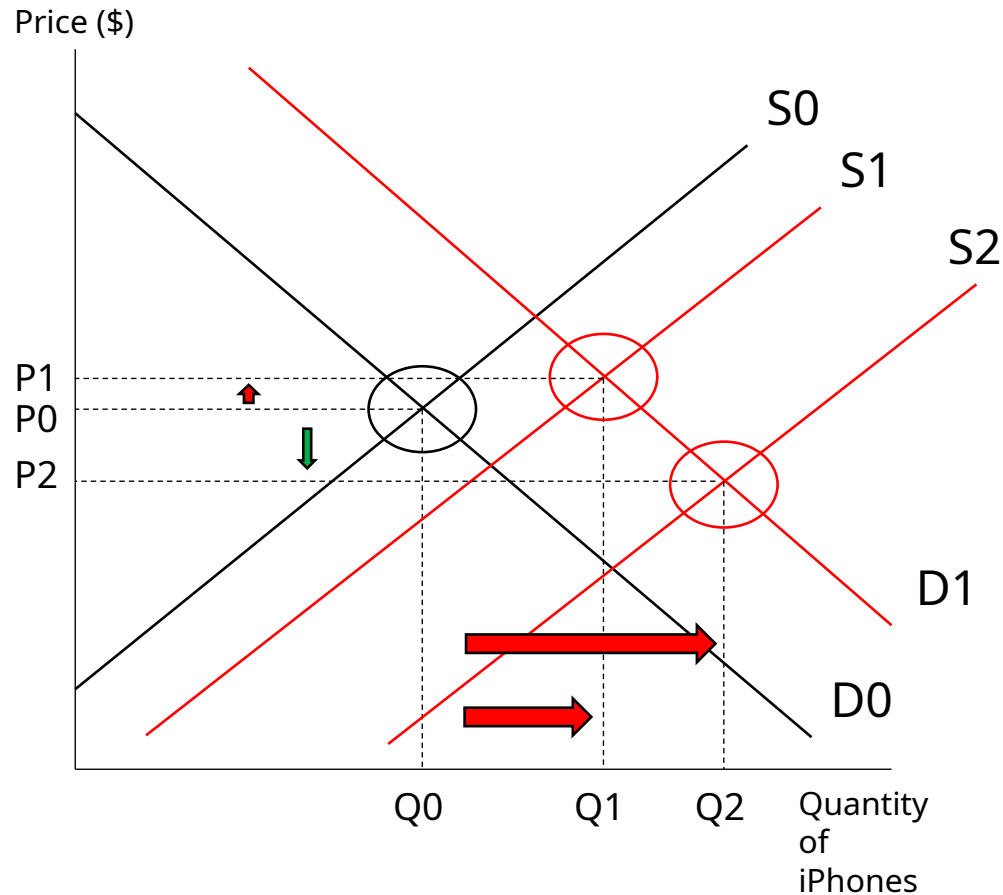
$$Q_D = 80$$

$$Q_S = 50$$

$$Q_D > Q_S$$

Shortage

Simultaneous Shifts in Demand and Supply: Increase in Demand and Supply



Our market is in equilibrium at price, P_0 , and quantity, Q_0 .

Suppose there is an increase in demand and supply, so that the new demand curve is D_1 and the new supply curve is S_1 .

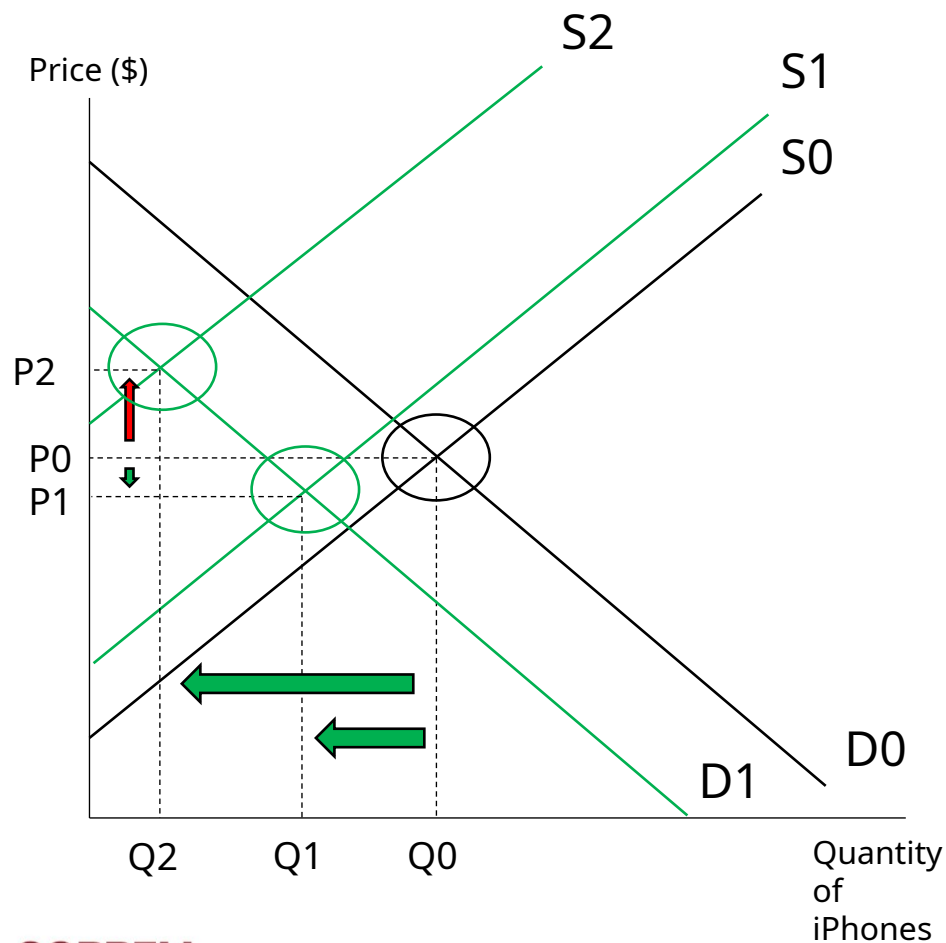
At the new equilibrium, price is P_1 and quantity is Q_1 .

In this scenario, quantity increased as expected, and price increased as well.

Suppose that supply increased to S_2 . Quantity increases to Q_2 , but now price falls below the original equilibrium price to P_2 .

Economists call price indeterminate because we don't know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

Decrease in Demand and Supply



Our market is in equilibrium at price, P_0 , and quantity, Q_0 .

Suppose there is a **decrease** in demand and supply, so that the new demand curve is D_1 and the new supply curve is S_1 .

At the new equilibrium, price is P_1 and quantity is Q_1 .

In this scenario, quantity decreased as expected, and price increased as well.

Suppose that supply instead decreased to S_2 . Quantity increases to Q_2 , but now price rises above the original equilibrium price to P_2 .

Economists call price indeterminate because we don't know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

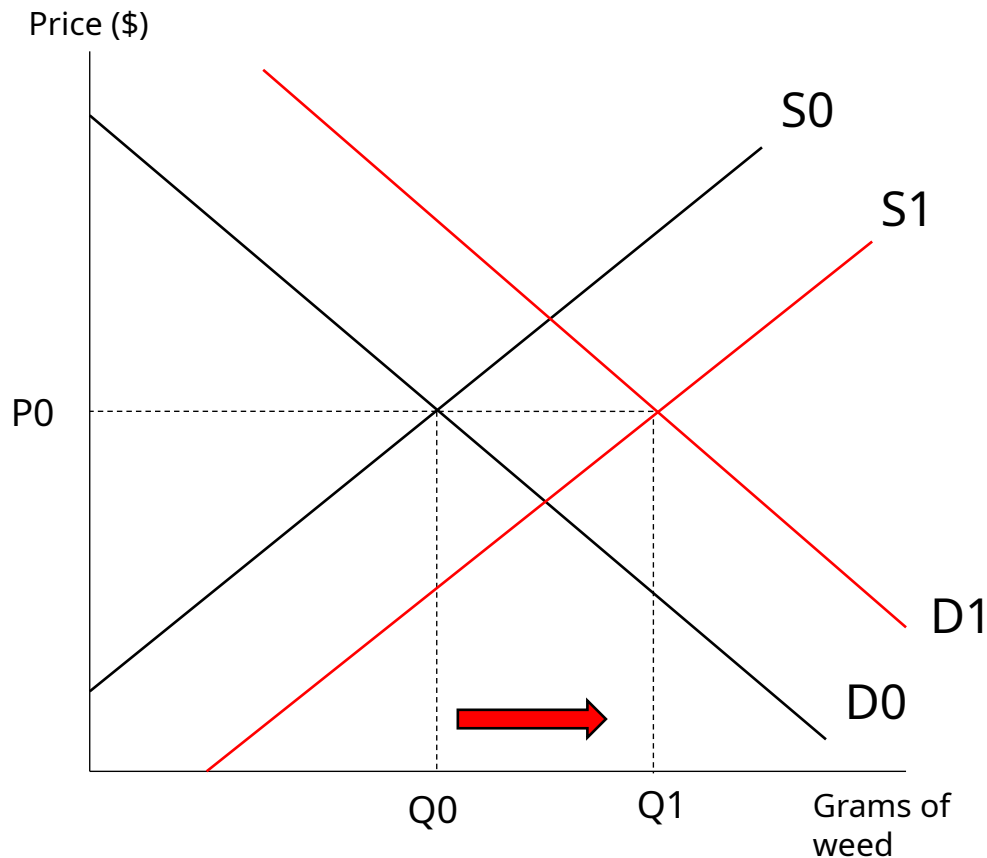
Simultaneous Shifts in Demand and Supply

Demand	Supply	Price	Quantity
Increase	Increase	Indetermina nt	Increase
Increase	Decrease	Increase	Indetermina nt
Decrease	Decrease	Indetermina nt	Decrease
Decrease	Increase	Decrease	Indetermina nt

Discussion Activity 1

- Suppose gas prices rise sharply. What happens to the market for used SUVs? Why?
- Think about substitution and cost changes. Does supply or demand shift?

Policy Change – Marijuana Legalization



What is the effect of a legalization of marijuana?

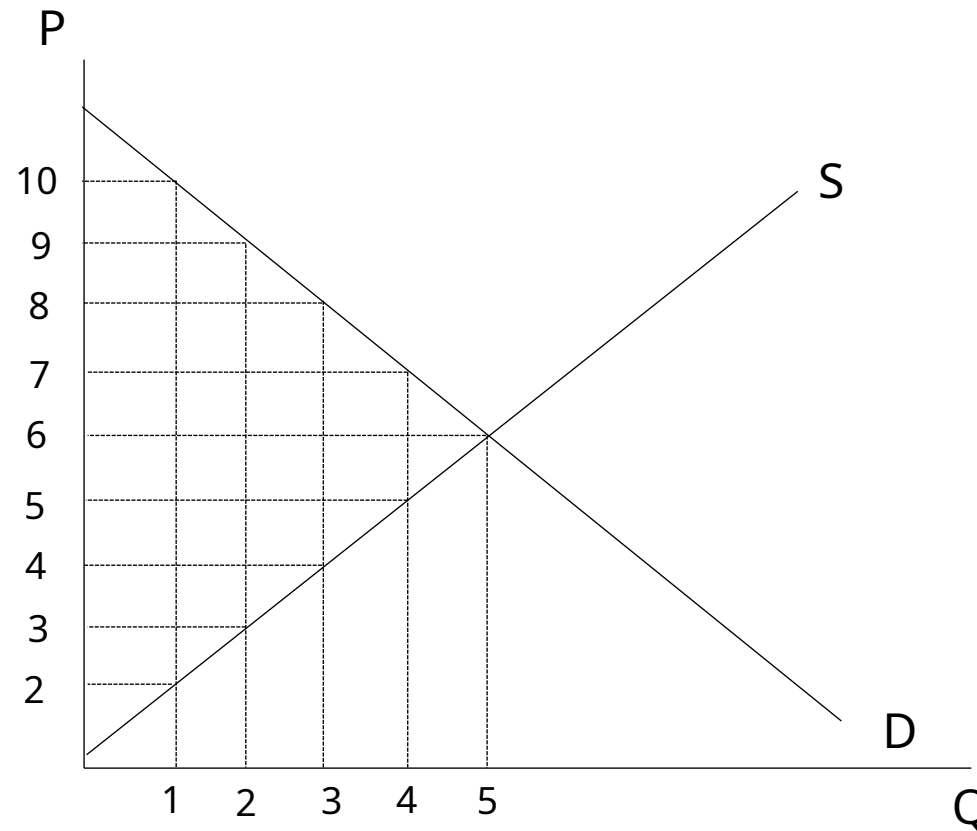
Demand and supply both **increase**. Quantity increases, but price is indeterminate.

More people will begin smoking weed and more people will begin to sell weed.

We know quantity increases for both supply and demand, but we don't know what happens to price even though a rise in price is depicted.

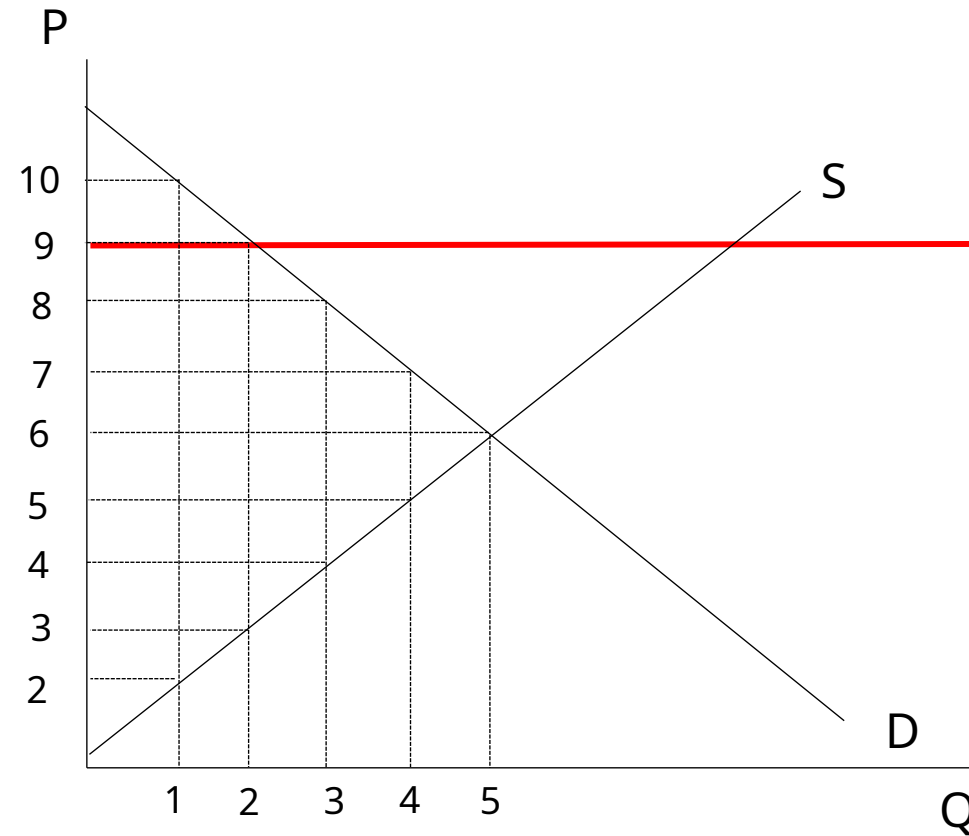
Market Makers

Suppose there are nine sellers and nine buyers, each willing to buy or sell one unit of a good, with values $\{\$10, \$9, \$8, \$7, \$6, \$5, \$4, \$3, \$2\}$. Assuming no transactions costs and a competitive market, what is the equilibrium price in this market?



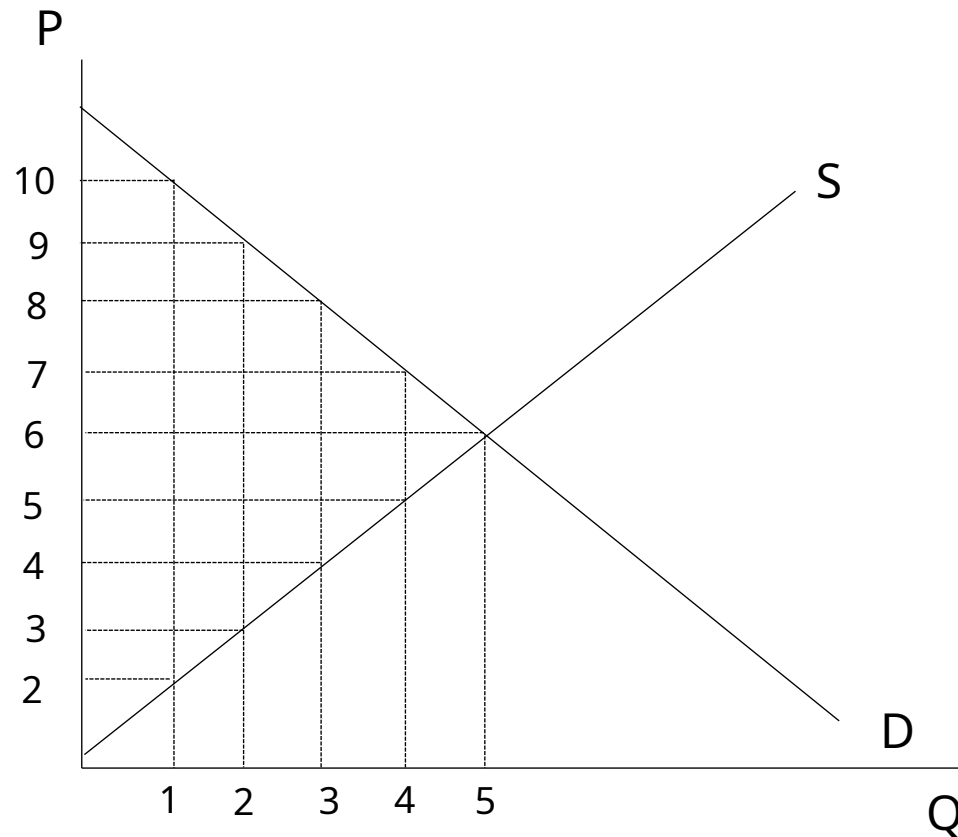
Market Makers cont.

If the government imposes a price floor at \$9 (i.e., price must be \$9 or higher) in the above market, how many goods will be traded?



Market Makers cont.

Suppose there is a single market maker in this market.
What is the optimal bid-ask spread?



Bid	Q	Ask	Q	Cost	Profit
2	1	10	1	\$2	$\$10 - \$2 = \$8$
4	3	8	3	\$12	$\$24 - \$12 = \$12$
5	4	7	4	\$20	$\$28 - \$20 = \$8$
6	5	6	5	\$30	$\$30 - \$30 = 0$

Discussion Activity 2

- If a government sets a price floor above equilibrium (e.g., minimum wage or crop price), who benefits and who loses?
- Think about producers who want to sell more vs. consumers who buy less.

Knowledge Check Answers

1. Movement: price change; Shift: change in factors like income, preferences
2. Surplus — too much supply, not enough demand
3. Population, income, preferences.
4. Number of sellers, technology, input prices, taxes/subsidies
5. It depends on the magnitude of each shift — direction of price change isn't always clear

Key Takeaways

- A market is defined by its product, geography, and time, define it clearly before analysis.
- Demand reflects buyers' willingness and ability to purchase; supply reflects sellers' willingness and ability to sell.
- Law of demand: as price falls, quantity demanded rises (and vice versa); law of supply: as price rises, quantity supplied rises.
- Market equilibrium occurs where supply equals demand, no surplus or shortage.
- Shifts in supply or demand result from external changes (e.g., income, technology, input costs).
- Simultaneous shifts can increase or decrease quantity, but price is often indeterminate.
- Surpluses result from prices too high; shortages from prices too low.
- Market makers reduce transaction costs; bid-ask spreads reflect cost of making a market.
- Government policies (e.g., price floors) can distort markets and reduce efficiency.

Knowledge Check

1. What causes movement along a demand curve vs. a shift of the curve?
2. What happens when the price is above the equilibrium level?
3. Name two factors that can shift the demand curve.
4. Name two factors that can shift the supply curve.
5. Why is price sometimes indeterminate when both supply and demand shift?

Consultant's Challenge: Diagnosing a Market Misfit

Scenario:

- You're advising a specialty bakery that launched a new gluten-free product line. Despite high reviews, sales are weak. The owner thinks it's just "bad luck."

Your Task:

1. Collaborate with 1-2 other of your classmates to answer these questions.
2. How would you use supply and demand analysis to identify what's really happening in their market?
3. What shifts in demand or supply might be impacting sales and how should the business respond?